

Homework (Jalapeno)

- Q1.** The absolute value of x is defined as the distance from x to:
- (A) 1
 - (B) 0
 - (C) -1
 - (D) x^2
 - (E) $|x|$
- Q2.** The absolute value of -5 is:
- (A) -5
 - (B) 5
 - (C) -5 and 5
 - (D) $|5|$
 - (E) None
- Q3.** In ancient Egypt, a pharaoh's treasure was hidden $|-8|$ meters below the surface. How many meters below the surface was it?
- (A) 6
 - (B) 8
 - (C) -8
 - (D) 10
 - (E) 12
- Q4.** The Mayans used a number system based on $|x| + 3$. If $x = -2$, what is $|x| + 3$?
- (A) 1
 - (B) 3
 - (C) 5
 - (D) 7
 - (E) 9
- Q5.** A historical artifact was found $|2x - 5|$ kilometers from a major city, where $x = 3$. How many kilometers was it from the city?
- (A) 1
 - (B) 3
 - (C) 5
 - (D) 7
 - (E) 9
- Q6.** What is $|0 - 4|$?
- (A) -4
 - (B) 0
 - (C) 4
 - (D) -2
 - (E) 2
- Q7.** A timeline of historical events spans $|x - 3|$ centuries, where $x = 7$. How many centuries does it span?
- (A) 2
 - (B) 4
 - (C) 6
 - (D) 8
 - (E) 10
- Q8.** The ancient Greeks used a unit of measurement called $|x| \cdot 2$, where $x = -3$. What is $|x| \cdot 2$?
- (A) 4
 - (B) 6
 - (C) 8
 - (D) 10
 - (E) 12

Open Ended Questions (FRQ)

- Q9.** A legacy of the ancient Romans includes the construction of roads $|x - 2|$ kilometers long, where x varies. Evaluate $|x - 2|$ if $x = -5$. Show your work and explain the historical significance of this road system. Required space:
- Q10.** The famous Rosetta Stone was discovered $|2x + 5|$ meters from the surface, where $x = -2$. Calculate the distance from the surface and describe the importance of the Rosetta Stone in understanding ancient languages. Required space:

Chef's Tips

- Consider using historical timelines to illustrate absolute value concepts
- Use artifacts to create word problems that are contextually relevant
- Emphasize the importance of absolute value in real-world applications